

VIDYASAGAR UNIVERSITY

Midnapore, West Bengal



PROPOSED CURRICULUM&SYLLABUS (DRAFT) OF

**BACHELOR OF BUSINESS ADMINISTRATION-BBA
(HONOURS)**

4-YEAR UNDERGRADUATE PROGRAMME

(w.e.f. Academic Year 2023-2024)

Based on

**Curriculum & Credit Framework for Undergraduate
Programmes (CCFUP), 2023& NEP, 2020**

VIDYASAGAR UNIVERSITY
BACHELOR OF BUSINESS ADMINISTRATION- BBA (HONOURS)
(under CCFUP, 2023)

Level	YR.	SEM	Course Type	Course Code	Course Title	Credit	L-T-P	Marks		
								CA	ESE	TOTAL
BBA (Hons.)	1 st	I	SEMESTER-I							
			Major-1	BBAH MJ101	T: Principles of Management	4	3-1-0	15	60	75
			SEC	BBASEC01	P: Computer Applications in Business	3	0-0-3	10	40	50
			AEC	AEC01	Communicative English -I	2	2-0-0	10	40	50
			MDC	MDC01	Social Value and Ethics	3	3-0-0	10	40	50
			VAC	VAC01	ENVS (<i>common for all programmes</i>)	4	2-0-2	50	50	100
			Minor-1	BBAM I01	T: Managerial Economics	4	3-1-0	15	60	75
			Semester-I Total							
		II	SEMESTER-II							
			Major-2	BBAH MJ102	T: Organisational Behaviour	4	3-1-0	15	60	75
			SEC	BBASEC02	P: Industrial Visit	3	0-0-3	10	40	50
			AEC	AEC02	E-Commerce	2	2-0-0	10	40	50
			MDC	MDC02	Digital Technologies	3	3-0-0	10	40	50
			VAC	VAC02	Yoga Wellness	4	4-0-0	10	40	50
			Minor-2	BBAM I02	T: Quantitative Techniques for Business Decisions	4	3-1-0	15	60	75
			Summer Intern.	CS	Community Service	4	0-0-4	-	-	50
			Semester-II Total							
			TOTAL of YEAR-I							

MJ = Major, MI = Minor Course, SEC = Skill Enhancement Course, AEC = Ability Enhancement Course, MDC = Multidisciplinary Course, VAC = Value Added Course; CA = Continuous Assessment, ESE = End Semester Examination, T = Theory, P = Practical, L-T-P = Lecture-Tutorial-Practical, MIL = Modern Indian Language, ENVS = Environmental Studies

MAJOR (MJ)

MJ-1: PRINCIPLES OF MANAGEMENT

Credits 04(Full Marks: 75)

OBJECTIVE OF THE COURSE

The course provides an overview of management and its evolution. It examines management functions of planning, organizing, leading, and controlling and its impact on the business organization.

Learning Objectives:

1. To explain the evolution of Management and its principles.
2. To discuss the functions of management and their importance in business.
3. To propose the application of the principles of management in an organizational setup.

MJ-1T: Principles of Management

Credits 04

Unit-I: Management-Meaning, Nature & Significance-Combination of Art & Science, Management as a Profession, Management Vs Administration, Levels of Management-Elements of managerial processes-Styles & Roles of Managers in Organizations. Contributions of Taylor and Fayol, Human Relations & Behavioural Schools-Hawthorne Studies.

Unit-II: Planning-Nature, Process of Planning, Planning and Environmental Uncertainties, Types of Planning, Advantages and Limitations of Planning-Decision Making-Stages in Decision Making.

Unit-III: Nature & Significance of Organization, Authority & Responsibility Relationships-Span of Control, Process of Delegations-Barriers to Delegation, Centralization & Decentralization. Concept of Line & Staff-Overcoming Line-staff conflict, Committees, Co-ordination, Organization Structures, Types, Advantages & Disadvantages.

Unit-IV: Staffing, Motivation & Leadership. Scope of Staffing Functions, Theories of Motivation. Theory X, Theory Y, Theory Z. Maslow's need hierarchy. Leadership Styles.

Unit-V: Control – Concept, Types, Importance, Process, Barriers to effective control and remedies, Techniques of control.

Suggested Readings:

1. Koontz & Heinz Weihrich: Essential of Management, McGraw Hill.
2. Richard L. Daft: Principles of Management, Cengage Learning India.
3. Stephen P. Robbins & Mary Coulter: Management, Pearson.
1. Stoner & Wankel: Management, Prentice Hall of India.

SKILL ENHANCEMENT COURSE (SEC)

SEC 1: Computer Applications in Business

Credits 03(Full Marks: 50)

OBJECTIVE OF THE COURSE -

Business Computer Applications focuses on integrating computer technology with decision-making and problem-solving skills. Areas of instruction include advanced applications in word processing, spreadsheets, presentation and database software that prepare students for industry readiness.

Learning Objectives:

- To understand and get practical exposure to MS Office
- To understand the application of MS Office in business scenarios

SEC 1P: Computer Applications in Business

Credits 03

Course Outline:

Unit I: Microsoft Word- Functionalities in the Microsoft Word; Creation of a new document; Editing an document; Saving and printing a file; Use of the different tools; Handling tables and images in MS Word, Printing an MS Word Document.

Unit II: Microsoft Excel- Creating a new spreadsheet document and editing an existing document; Mathematical calculations (SUM, AVG, MAX, MIN, etc.) using MS Excel; Reference Operators, Functions & Formulae; Sorting & Filtering; Creating a column/ bar chart; Changing the size and position of a chart; Data validation.

Unit III: Microsoft Access- Creating New and Opening Existing Databases; Creating a database using a wizard and without using a wizard; Tables –Introduction and advanced features; Retrieving data using SQL query; Record manipulation using Forms and Reports.

Unit IV: Microsoft PowerPoint- Functionalities of a PowerPoint presentation; Creating and inserting a new slide; Creating a title slide and other slide layouts; Applying a design template; Using a two-column text; Applying basic transition and animation; Running the slide show, Printing the slides.

Suggested Readings:

1. Leon, A., Leon, M., Introduction to Computers (1st Edition), Vikas Publishing
2. Ram, B., Computer Fundamentals: Architecture and Organization, New Age International
3. Script Demics, Learn to Master Microsoft Office 2016/365: 4 Color Book, Staredu Solutions India
4. Laudon, J.P., Laudon, K.C., Management Information Systems (16th Edition), Pearson
5. Ramirez, A., Excel Formulas and Functions 2020: The step-by-step Excel Guide with Examples on How to Create Powerful Formulas, independently published
6. Singh, I., Learn to use computer, MS Word, PowerPoint and Excel (1st Edition), independently published

Ability Enhancement Course (AEC)

Ability Enhancement Course (AEC) - 01

AEC-1T: Communicative English-01

Credits 02 (Full Marks-50)

Course Contents:

1. Communication Skills

- a) Types and Models of Communication
- b) Verbal and Non-verbal Communication
- c) Barriers and Strategies
- d) Inter-personal Communication

2. Listening Skills:

- a) Active and Passive Listening

3. Speaking Skills:

- a) Different forms of Speaking – Formal/Informal
- b) Group Discussion
- c) English in Situations: Greeting & Leave Taking, Making & Granting/Refusing Requests, Queries & Giving Information/Direction, Describing objects/process, Narrating events & Commentary, Persuasion & Motivation, Complaints & Apologies, Expressing disapproval, Alerting & Warning

4. Reading Skills:

- a) Different types of Reading
- b) Comprehension

Suggested Readings:

- 1. *A Textbook of English and Communication Skills*. Richa Mishra and Ratna Rao, New Delhi: Macmillan: 2019
- 2. *Many Coloured Glass*. Board of Editors, Department of English, Vidyasagar University, Delhi: Macmillan, 2013.
- 3. *Connect: Course in Communicative English*, Debashis Bandyopadhyay & Malathi Krishnan, Cambridge University Press, New Delhi, 2018.

Course contents:

Unit-I: Philosophy of Life and Individual Qualities

Human Life on Earth - Purpose of Life, Meaning and Philosophy of Life. The Law of Nature – Protecting Nature /Universe. Basic Culture - Thought Analysis - Regulating desire - Guarding against anger - To get rid of Anxiety – The Rewards of Blessing - Benevolence of Friendship - Love and Charity - Self – tranquility/Peace.

Unit-II: Social Values (Individual and Social Welfare)

Family - Peace in Family, Society, The Law of Life Brotherhood - The Pride of Womanhood – Five responsibilities/duties of Man : - a) to himself, b) to his family, c) to his environment, d) to his society, e) to the Universe in his lives, Thriftiness (Thrift)/Economics. Health - Education - Governance - People's Responsibility / duties of the community, World peace.

Unit-III: Mind Culture & Tending Personal Health

Mind Culture - Life and Mind - Bio - magnetism, Universal Magnetism (God –Realization and Self Realization) - Genetic Centre – Thought Action – Short term Memory – Expansiveness – Thought – Waves, Channelizing the Mind, Stages - Meditation, Spiritual Value. Structure of the body - the three forces of the body- life body relation, natural causes and unnatural causes for diseases, Methods in Curing diseases.

Suggested Readings:

1. Value Education for Health, Happiness and Harmony, The World Community Service, Centre Vethathiri Publications (Unit 1 – III).
2. Thirukkural with English Translation of Rev. Dr. G.U. Pope, Uma Publication, 156, Serfoji Nagar, Medical College Road, Thanjavur 613 004 (for Units I - III).
3. R S Nagaarazan, Textbook on Professional Ethics And Human Values, New Age International Publishers, 2006 (for Units IV-V).
4. Bharatiya Dharmaneeti Edited by Amita Chatterjee, published by Jadavpur University; Dvitiya samskarana edition (2013)

Value Added Course (VAC)

Value Added Course (VAC)- 01

VAC-01: ENVS- Environmental Studies

Credits 04 (Full Marks-100)

Course contents:

Unit-I: Introduction to environmental studies

(2 lectures)

- Multidisciplinary nature of environmental studies;
- Scope and importance; Concept of sustainability and sustainable development.

Unit-II: Ecosystems

(6 lectures)

- What is an ecosystem? Structure and function of ecosystem; Energy flow in an ecosystem: food chains, food webs and ecological succession. Case studies of the following ecosystems: a) Forest ecosystem; b) Grassland ecosystem c) Desert ecosystem d) Aquatic ecosystems (ponds, streams, lakes, rivers, oceans, estuaries)

Unit-III: Natural Resources: Renewable and Non-renewable Resources

(8 lectures)

- Land resources and land use change; Land degradation, soil erosion and desertification;
- Deforestation: Causes and impacts due to mining, dam building on environment, forests, biodiversity and tribal populations. Joint forest management.
- Water: Use and over--exploitation of surface and ground water, floods, droughts, conflicts over water (international & interstate).
- Energy resources: Renewable and non renewable energy sources, use of alternate energy sources, growing energy needs, case studies.

Unit-IV: Biodiversity and Conservation

(8 lectures)

- Levels of biological diversity: genetic, species and ecosystem diversity; Bio-geographic zones of India; Biodiversity patterns and global biodiversity hot spots
- India as a mega--biodiversity nation; Endangered and endemic species of India
- Threats to biodiversity: Habitat loss, poaching of wildlife, man--wildlife conflicts, biological invasions; Conservation of biodiversity: In-situ and Ex-situ conservation of biodiversity.
- Ecosystem and biodiversity services: Ecological, economic, social, ethical, aesthetic and Informational value.

Unit-V: Environmental Pollution

(8 lectures)

- Environmental pollution: types, causes, effects and controls; Air, water, soil and noise pollution
- Nuclear hazards and human health risks
- Solid waste management: Control measures of urban and industrial waste.
- Pollution case studies.
- Noise pollution.

Unit-VI: Environmental Policies & Practices

(7 lectures)

- Climate change, global warming, ozone layer depletion, acid rain and impacts on human communities and agriculture
- Environment Laws: Environment Protection Act; Air (Prevention & Control of Pollution) Act; Water (Prevention and control of Pollution) Act; Wildlife Protection Act; Forest Conservation Act. International agreements: Montreal and Kyoto protocols and Convention on Biological Diversity (CBD).
- Nature reserves, tribal populations and rights, and human wildlife conflicts in Indian context.
- Environmental policy and gender issues

Unit-VII: Human Communities and the Environment

(6 lectures)

- Human population growth: Impacts on environment, human health and welfare.
- Resettlement and rehabilitation of project affected persons; case studies.
- Disaster management: floods, earthquake, cyclones and landslides.
- Environmental movements: Chipko, Silent valley, Bishnois of Rajasthan.
- Environmental ethics: Role of Indian and other religions and cultures in environmental conservation.
- Environmental communication and public awareness, case studies (e.g. CNG vehicles in Delhi)

Unit-VIII: Field work

(Equal to 5 lectures)

- Visit to an area to document environmental assets: river/ forest/ flora/fauna, etc.
- Visit to a local polluted site--Urban/Rural/Industrial/Agricultural.
- Study of common plants, insects, birds and basic principles of identification.
- Study of simple ecosystems--pond, river, Delhi Ridge, etc.
- Disaster management.
- Coastal ecosystem

Suggested Readings:

- i. Carson, R. 2002. *Silent Spring*. Houghton Mifflin Harcourt.
- ii. Gadgil, M., & Guha, R. 1993. *This Fissured Land: An Ecological History of India*. Univ. of California Press.
- iii. Gleeson, B. and Low, N. (eds.) 1999. *Global Ethics and Environment*, London, Routledge.
- iv. Gleick, P. H. 1993. *Water in Crisis*. Pacific Institute for Studies in Dev., Environment & Security. Stockholm Env. Institute, Oxford Univ. Press.
- v. Groom, Martha J., Gary K. Meffe, and Carl Ronald Carroll. *Principles of Conservation Biology*. Sunderland: Sinauer Associates, 2006.
- vi. Grumbine, R. Edward, and Pandit, M.K. 2013. Threats from India's Himalaya dams. *Science*, 339: 36--37.
- vii. McCully, P. 1996. *Rivers no more: the environmental effects of dams* (pp. 29--64). Zed Books.
- viii. McNeill, John R. 2000. *Something New Under the Sun: An Environmental History of the Twentieth Century*.
- ix. Odum, E.P., Odum, H.T. & Andrews, J. 1971. *Fundamentals of Ecology*. Philadelphia: Saunders.
- x. Pepper, I.L., Gerba, C.P. & Brusseau, M.L. 2011. *Environmental and Pollution Science*. Academic Press
- xi. Rao, M.N. & Datta, A.K. 1987. *Waste Water Treatment*. Oxford and IBH Publishing Co.Pvt. Ltd.
- xii. Raven, P.H., Hassenzahl, D.M. & Berg, L.R. 2012. *Environment*. 8th edition. John Wiley & Sons.

MINOR (MI)

MI – 1: Managerial Economics

Credits 04(Full Marks: 75)

OBJECTIVE OF THE COURSE

This course is designed to provide the students with the fundamental knowledge about micro and macroeconomic concepts and theories that are commonly applied to facilitate various commercial decision-making processes for the company managers and administrators.

Learning Objectives:

- To understand key microeconomic concepts to understand consumers, production, and markets.
- To understand the different types of market and their characteristics.
- To understand macroeconomic business environment particularly related to India.

MI – 1T: Managerial Economics

Credits 04[Theory: 60 Hrs]

Course contents:

Unit-I: Economic terms and basic concepts, Economic Problems (Problem of Scarcity, Problem of Choice, Problem of Efficiency), Production Possibility Frontier, Definition of Managerial Economics, Nature and Scope of Managerial Economics, Comparison/relationship between Managerial Economics and other allied subjects such as Mathematics, Statistics, Accounting, and Management.

Unit-II: Demand, Supply and Market equilibrium: individual Demand, market demand, individual supply, market supply, market equilibrium, Elasticities of demand and supply: Price elasticity of demand, income elasticity of demand, cross price elasticity of demand, Factors determining demand elasticity, elasticity of supply. Theory of consumer behavior: cardinal utility theory, ordinal utility theory, Consumer Equilibrium based on indifference curve and budget line.

Unit-III: Theory of Production: Law of variable proportion, Concepts of Total Product (TP), Average Product (AP), and Marginal Product (MP), Relationship between TP, AP and MP, three stages of production. Theory of costs: Fixed and variable costs, Economies of scale and long run costs, Economies of scope, Total Revenue and Profit, Producer optimization in the short run (Profit Maximization with Isoquant and Isocost line).

Unit-IV: Theory of firm and market organization: Perfect Competition (basic features, short run equilibrium of firm/industry, long run equilibrium of firm/industry), Monopoly (basic features, short run equilibrium, long run equilibrium, comparison with perfect competition), Price discrimination, Monopolistic Competition (basic features and long run equilibrium), Basic concept of Oligopoly market.

Unit-V: Basic concepts of Macroeconomics and their relevance in managerial decision making, National Income; Measurement of National Income and GDP, Circular flow of National Income, Monetary and Fiscal Policy, Inflation control.

Suggested Readings:

1. Dominick Salvatore (2009). Principles of Microeconomics (5th ed.) Oxford University Press
2. Lipsey and Chrystal. (2008). Economics. (11th ed.) Oxford University Press
3. Koutosyannis (1979). Modern Micro Economics. Palgrave Macmillan

MJ-2:Organisational Behaviour

Credits 04(Full Marks: 75)

OBJECTIVE OF THE COURSE

To provide knowledge about Organizational Behaviour, individual and group behaviour and give an overview about change in organization and QWL.

MJ-1T:OrganisationalBehaviour

Credits 04 [Theory: 60 Hrs]

UNIT-I Introduction: Nature and scope of OB, Challenges and opportunities for OB, Organization Goals, Models of OB, Impact of Global and Cultural diversity on OB.

UNIT-II Individual Behavior: Concept, Personality, Perception and its role in individual decision making, Learning, Motivation: Hierarchy of needs theory, Theory X and Y, Motivation Hygiene theory, Vroom's expectancy theory.

UNIT-III Group Dynamics: Definition and classification of Groups, Formation of Groups; Teams Vs Groups, Group Structures, Group decision making, Contemporary issues in managing teams, Inter-group problems in organizational group dynamics, Management of conflict.

UNIT-IV Management of Change: Change and Organizational development, Resistance to change, Approaches to managing organizational change, Organizational effectiveness, Organizational culture, Power and Politics, Stress Management: Definition, potential, sources of stress, consequences of stress, managing stress.

Suggested Readings:

1. Robbins, Judge, and Vohra (2013); Organizational Behavior; Prentice Hall Inc.; 15th Edition.
2. Fred, Luthans (2005); Organisational Behavior; UK: McGraw Hill; 10th Edition.
3. G, Moorhead & Griffith. (2007). Organizational Behavior. Houghton Muffin Co.
4. Newstrom J (2007); OB: Human Behaviour at Work. McGraw Hill Inc.; 12th Edition.
5. PareekUdai (2016); Understanding Organizational Behaviour; Oxford University Press; 4th Edition.
6. Rao, V.S.P. (2009); Organization Behaviour; Excel Book.
6. Banerjee R P (2023): Stress Management through Mind Engineering, Routledge, Taylor and Francis Group.

SEC 2: Industrial Visit

Credits 03

OBJECTIVE OF THE COURSE -

Objective of industrial visit is to provide student an insight regarding internal working of companies. Theoretical knowledge is not enough for making a good professional career. With an aim to go beyond academics, industrial visit provides student a practical perspective on the world of work.

SEC 2P: Industrial Visit

Full Marks: 50

Course Outline:

After industrial visit, students are supposed to submit a report containing his /her personal observation and realization. The students need to submit

- Summary of the Report
- Case report presentation through Power-Point presentation, keeping in view his /her personal observation and realization.

25marks has been assigned for Report writing and it should be evaluated by external examiners and remaining 25 marks has been assigned for presentation of Report before the internal and external examiners.

ABILITY ENHANCEMENT COURSE (AEC)

AEC 2: E-Commerce

Credits 02 (Full Marks: 50)

OBJECTIVE OF THE COURSE -

The objectives of this course are to provide student with a comprehensive understanding of electronic commerce and its significance in modern business practices.

Course contents:

Unit I: Introduction to E-commerce:

A Brief History of E-commerce, Understanding organizing themes, E-commerce Business Models, Major Models of E-commerce- Business to Consumer (B2C), Business to Business (B2B),

Unit II: E-commerce web site, Security and payment:

A systematic Approach of E-commerce, The E-commerce security environment, Security threats in the E-commerce environment, Technology solution, Management policies, Business procedures, and public laws, Payment system, E-commerce payment system, Electronic billing presentment and payment.

Unit III: E-commerce marketing concepts, Online retailing and services:

The Internet Audience and Consumer Behaviour, Basic Marketing Concepts, Internet Marketing Technologies, B2C and B2B E-commerce marketing and business strategies, The Retail sector, Analyzing the viability of online firms, E-tailing Business Models, Common Themes in online retailing, The service sector -offline and online, online financial services, Online Travel Services, Online career services.

Suggested Readings:

1. Kenneth C. Laudon, E-Commerce: Business, Technology, Society, 4th Edition, Pearson.
2. S. J. Joseph, E-Commerce: An Indian perspective, PHI.
3. Elias. M. Awad, " Electronic Commerce", Prentice-Hall of India Pvt Ltd.

Course contents:

Unit-I: Introduction and Evolution of Digital systems, Role and significance of Digital Technology, Information & communication technology & tools, Computer system & it's working, Software and its types, Operating Systems: types and functions.

Communication systems: Principles, model & transmission media, Computer networks, Internet: concept and applications, WWW, Web Browsers, search engines, Messaging, e-mail, social networking.

Unit- II: Computer Based Information system: significance and types, e-Commerce & digital marketing: basic concepts, benefits & challenges.

Digital India & e-Governance: Initiatives, Infrastructure, Services and Empowerment.

Digital financial tools: Unified Payment Interface, Aadhaar enabled payment System, USSD, Credit/Debit Cards, e-Wallets, Internet banking, NEFT/RTGS and IMPS, Online Bill Payments and PoS, Cyber Security: Threats, Significance, Challenges, Precautions, safety Measures & Tools,

Unit- III: Emerging Technologies & their applications: Overview of Cloud Computing, Big Data, Internet of things, Virtual reality, Block chain, robotics, Artificial intelligence, 3D Printing, Future of digital technologies.

Suggested Readings:

1. Fundamentals of Computers by E Nalagurusamy, ata mcgraw Hill.
2. Data Commination and Networking by Behrouz A. Forouzan, Mcgraw Hill Educaition.
3. Emerging Technologies in Computing: Theory, Practice, and Advances, by P.Kumar, A omar, and R.Sharmila, 1st Edition, 2021.
4. Essentials of cloud computingby K. Chandrasekharan, CrC press, 2014.
5. Blockchanin: Blueprint for a new economy by M. Swan, O'Reilly Media, 2015
6. Introduction to Computers by Peter Norton, Tata McGraw Hill.

Course contents:**Unit -I: Introduction of Yoga**

- 1.1 Meaning, Definition, Aim, Objectives and Importance of Yoga
- 1.2 History and Development of Yoga
- 1.3 Astanga Yoga, Hatha Yoga, Observation of International Yoga Day
- 1.4 Suryanamaskar, Pranayamas and Meditations.

Unit -II: Health and Health Problems in India

- 2.1 Health: Meaning, Definition, Dimensions and Factors.
- 2.2 Health Education: Meaning, Definition, Aim, Objectives and Principles. Personal Hygiene: Care of Eyes, Ear, Nose, Skin, Mouth, Teeth and Feet.
- 2.3 Aim, Objectives and Functions of Health Agencies: World Health Organization (WHO), United Nations Educational Scientific & Cultural Organization (UNESCO), United Nations International Children's Emergency Fund (UNICEF). Integrated Child Development Services (ICDS), Ministry of Health and Family Welfare (MHFW).
- 2.4 Life Style Diseases (Hypokinetic): obesity and Diabetes. Life Style Diseases (Hyperkinetic): Hypertension and Psychological Disorder-Stress.

Unit -III: Wellness and Wellness Programme

- 3.1 Wellness - Concept, Definition, Components, Significance with reference to Positive Lifestyle.
- 3.2 Concepts of Quality of Life and Body Image.
- 3.3 Factors affecting Wellness.
- 3.4 Wellness Programme in Reference to Physical Activities & Yoga

Suggested Readings

1. Corbin, C. B. G. J. Welk. W. R Corbin, K. A. Welk (2006) Concepts of Physical Fitness: for Wellness. McGraw Hill, New York, USA.
2. Kamlesh, M. L. & Singh, M. K. (2006) Physical Education (Naveen Publications).
3. Kansal, D.K. (2008) Textbook of Applied Measurement, Evaluation & Sports Selection. Sports & Spiritual Science Publications, New Delhi.
4. Uppal, A.K. (2004), Fitness and Health 5th ed. (U.K., Human Kinetics).
5. Sharma Jai Prakash And Sehgal Madhul(2006). Yog-Shiksha. Friends Publication. Delhi.
6. Mukerji, A.P. (2010). The Doctorine and Practice of Yoga, General Books, LLC, New Delhi.
7. Sarin N (2003). Yoga Dawara Rogoon Ka Upchhar. Khel Sahitya Kendra.
8. Text Book Hath Yoga Pradipika.
9. Text Book Patanjali Yoga Sutra.
10. Kayal, R. Yoga Sikha. Clasiqque Books. Kolkata.
11. Sahu, D. A Critical review on Modern Trend of Yogic Asana. Clasiqque Books. Kolkata.

4. Pindyck, Rubinfeld and Mehta. (2009). Micro Economics. (7th ed.). Pearson.
5. Vinita Agarwal: Managerial Economics (2013), Pearson.
6. Sampat Mukherjee: Business and Managerial Economics in Global Context (2009), NCBA

MI – 2: Quantitative Techniques for Business Decisions Credits 04(Full Marks: 75)

OBJECTIVE OF THE COURSE

Quantitative techniques can be understood as a collection of mathematical and statistical tools that are used to provide powerful means of analysis using quantitative data for effective decision making in business. These techniques involve systematics and scientific methods for solving complex problems for taking effective business decisions.

Learning Outcome: On completing the course students will be able to Understand and develop insights and knowledge base of various concepts of Quantitative Techniques. Develop skills for effectively analyze and apply Quantitative Techniques in decision making.

MI – 2T: Quantitative Techniques for Business Decisions Credits 04 [Theory: 60 Hrs]

Course contents:

1. **Basic Mathematical Tools:** Ratio & Proportion, Percentage, Profit & Loss, Simple & Compound Interest.
2. **Time and Number related Mathematical operations:** AP and GP Series, Number System, Permutation, Combination.
3. **Basic Statistics:** Meaning of statistics, scope and importance of statistics, classification and tabulation of data, presentation of data, graphical presentation of data by frequency curve, frequency polygon, ogives, histogram, limitations of statistics.
4. **Measures of Central Tendency:** Mean: Arithmetic (simple and weighted), Median – Mode; Standard Deviation, Variance etc.,
5. **Correlation and Regression Analysis:** Correlation: Significance of Correlation, Types of Correlation, Scatter Diagram Method, Karl Pearson Coefficient of correlation, Spearman's coefficient of Rank correlation. Regression: Introduction, Regression Lines and Coefficients.

Suggested Readings:

1. Goon Gupta and Das Gupta, Fundamentals of Statistics, Vol. 1, The World Press Pvt. Ltd., Kolkata.
2. Gupta and Kapoor, Fundamentals of Mathematical Statistics, Sultan Chand and Sons, New Delhi.
3. Praveen, R. V., Quantitative Aptitude and Reasoning, PHI Learning
4. Sharma, Arun, How to prepare for Quantitative Aptitude for the CAT, Tata McGraw Hill